EAST WEST UNIVERSITY

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

Data Communication Project Report

Topic: Data Communication in a network

Submitted By:

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Course: CSE 350 (Data Communication)

Section: 02

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Title: Data Communication in a Network using EIGRP with Three Routers and Three Switches

1.Problem Statement:

This project is to design and implement a network infrastructure using three routers and three switches, connecting six PCs (indexed from 1 to 6). The network will utilize the Enhanced Interior Gateway Routing Protocol (EIGRP) to ensure efficient and reliable routing. The PCs and network devices will be assigned IP addresses from a Class C IP address range. The final step will be to demonstrate successful communication (PING operation) between PC1, connected to the first router, and PC6, connected to the third router.

Objectives:

1. **Design a network topology using three routers and three switches.**
2. **Connect six PCs to the network, ensuring each is assigned a unique Class C IP address.**
3. **Configure the network using EIGRP to facilitate routing between the routers.**
4. **Demonstrate successful communication between PC1 and PC6 by performing a PING operation.**

Requirements:

1. **Hardware:**
    - **3 Routers**
    - **3 Switches**
    - **6 PCs**
    - **Appropriate cabling for connections**
2. **Software:**
    - **Network simulation software (e.g. Cisco Packet Tracer)**

3. **Network Protocol:**
    - **Enhanced Interior Gateway Routing Protocol (EIGRP)**
4. **IP Addressing:**
    - **Use Class C IP address range for all devices.**

Solution:

To design a network that includes three routers, three switches, and six PCs, we will adopt a hierarchical network design strategy. This approach involves segmenting the network into smaller subnetworks, each associated with a switch, and linking these switches to the routers. We will use the Enhanced Interior Gateway Routing Protocol (EIGRP) to facilitate efficient and reliable routing between the network segments.

Our network structure will ensure that each PC is connected to a switch, which in turn connects to a router. The routers will be configured to communicate using EIGRP, ensuring seamless data communication across the network. We will assign IP addresses from a Class C range to each router interface and PC, following a structured and organized addressing scheme.

For example, we will assign the following IP addresses to each router interface and PC:

- ☐ **PCs**
    - **PC 1: 192.168.1.11**
    - **PC 2: 192.168.1.12**
    - **PC 3: 192.168.2.11**
    - **PC 4: 192.168.2.12**
    - **PC 5: 192.168.3.11**
    - **PC 6: 192.168.3.12**

- ☐ **Router Interfaces**
  - **For Router 1:**

    Interface Fa 0/0 : 192.168.1.1

    Interface Fa 1/0 : 192.168.4.1

    Interface Fa 7/0 : 192.168.6.1

  - **For Router 2:**

    Interface Fa 0/0 : 192.168.2.1

    Interface Fa 1/0 : 192.168.4.2

    Interface Fa 6/0 : 192.168.5.1

  - **For Router 3:**

    Interface Fa 0/0 : 192.168.3.1

    Interface Fa 6/0 : 192.168.5.2

    Interface Fa 7/0 : 192.168.6.2

In this configuration, there will be six network segments, each with its own network address.

- ☐ **Network address:**

  192.168.1.0
  192.168.2.0
  192.168.3.0
  192.168.4.0
  192.168.5.0
  192.168.6.0

- **Network Gateways for Six PCs:**

  PC1, PC2 : 192.168.1.1

  PC3, PC4 : 192.168.2.1

  PC5, PC6 : 192.168.3.1

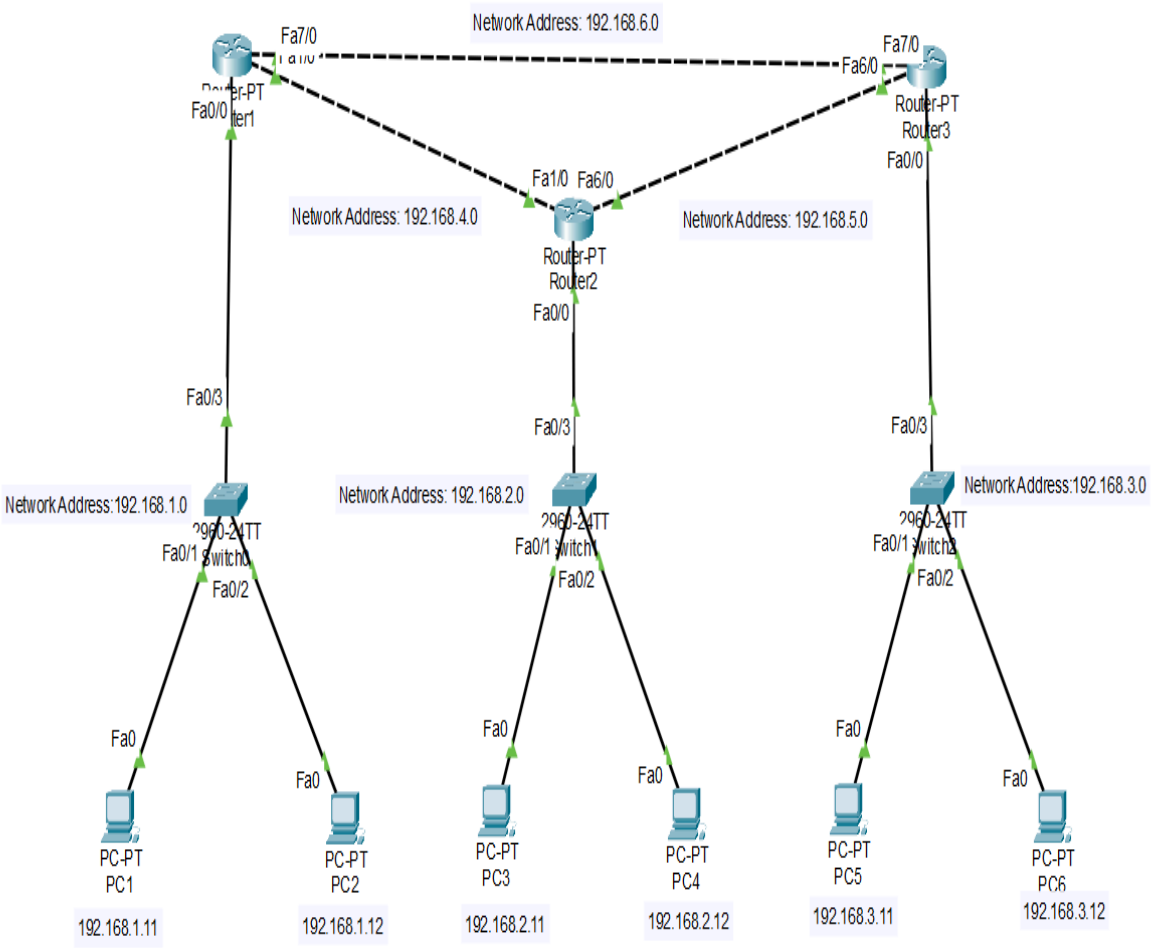
We will then configure the EIGRP protocol on all routers to enable routing between subnets. Each router will be set up with the network addresses of the subnets connected to its interfaces.

Once the network is configured, we will test the connection between PC1 on Router1 and PC6 on Router3 using the PING command. We will run the following command on PC1 to ping PC6:

```
ping 192.168.3.12
```

If the network is configured correctly, PC1 will receive a response from PC6.

2.Design:

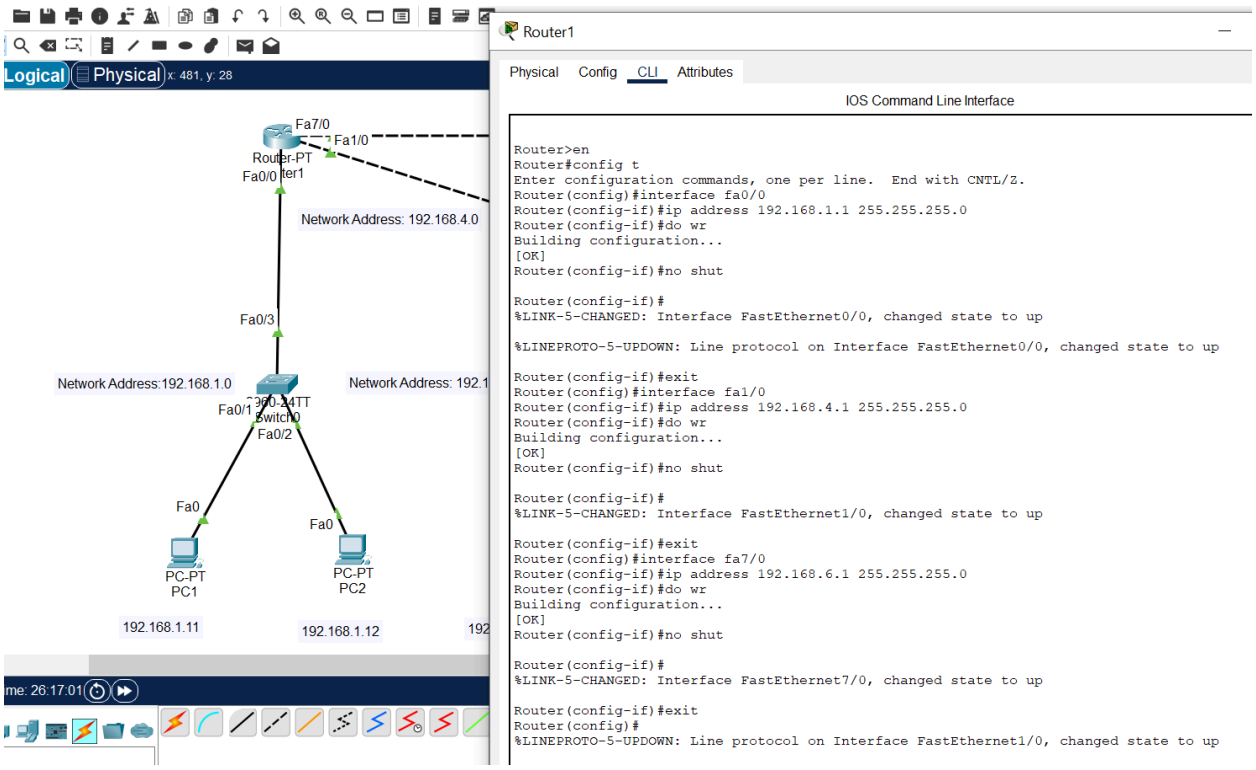

Network Address: 192.168.6.0
Fa7/0
Fa7/0
Fa6/0
Router-PT
Router3
Fa0/0
Fa1/0
Fa6/0
Network Address: 192.168.4.0
Network Address: 192.168.5.0
Router-PT
Router2
Fa0/0
Fa0/3
Fa0/3
Fa0/3
Network Address:192.168.1.0
Network Address: 192.168.2.0
Network Address:192.168.3.0
Switch0
Switch1
Switch2
Fa0/1
Fa0/2
Fa0/1
Fa0/2
Fa0/1
Fa0/2
Fa0
Fa0
Fa0
Fa0
Fa0
Fa0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5
PC-PT
PC6
192.168.1.11
192.168.1.12
192.168.2.11
192.168.2.12
192.168.3.11
192.168.3.12

- **Router Address Setup :**

Commands for Router's Setup :

Router 1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#interface fa1/0
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#interface fa7/0
Router(config-if)#ip address 192.168.6.1 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit
```



Router 2:

Router>enable

Router#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fa0/0

Router(config-if)#ip address 192.168.2.1 255.255.255.0

Router(config-if)#do wr

Building configuration...

[OK]

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#interface fa1/0

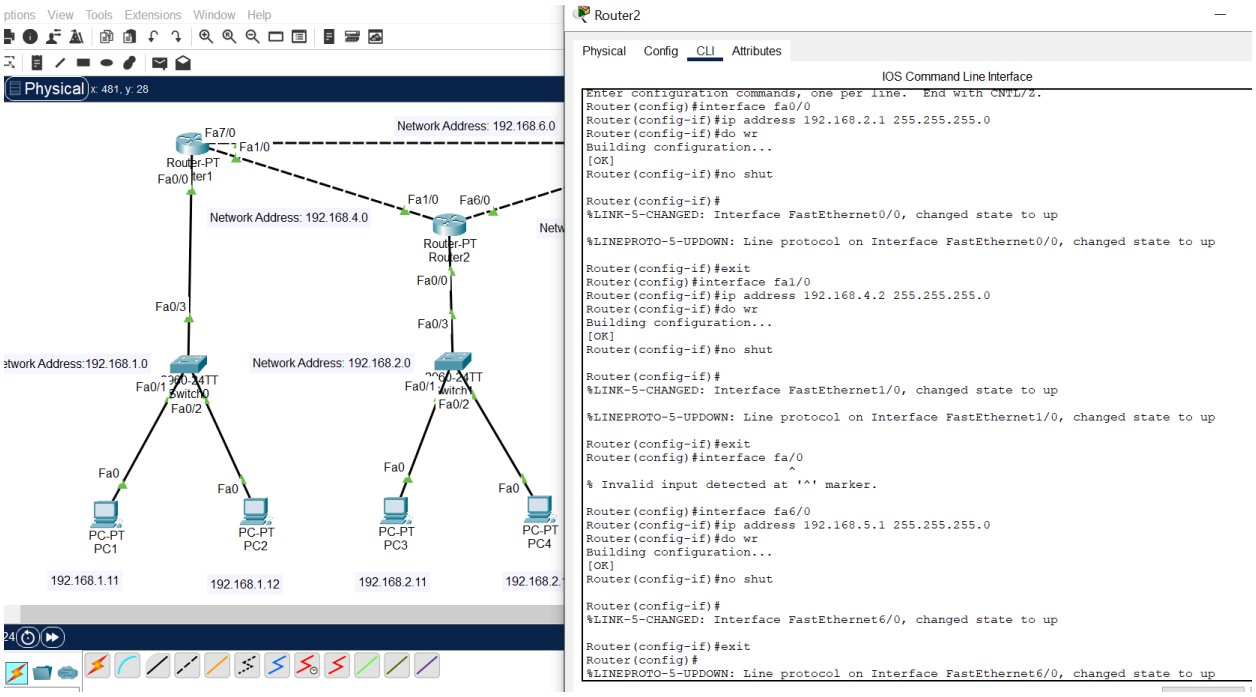
Router(config-if)#ip address 192.168.4.2 255.255.255.0

Router(config-if)#do wr

Building configuration...

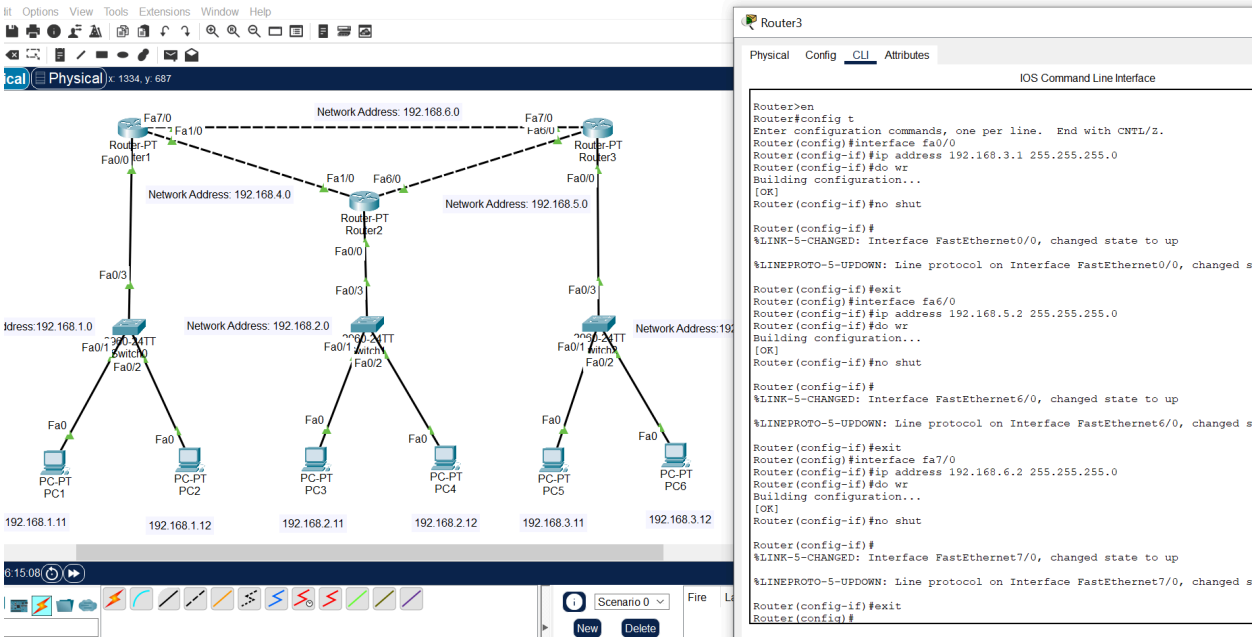
[OK]

Router(config-if)#no shut
Router(config-if)#exit
Router(config)#interface fa6/0
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit



Router 3:
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#do wr

Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#interface fa6/0
Router(config-if)#ip address 192.168.5.2 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#interface fa7/0
Router(config-if)#ip address 192.168.6.2 255.255.255.0
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#no shut
Router(config-if)#exit



- ☐ **Applying the EIGRP (Enhanced Interior Gateway Routing Protocol) to each router:**

We will use the following commands for each router:

Router 1:

Router>enable

Router#config

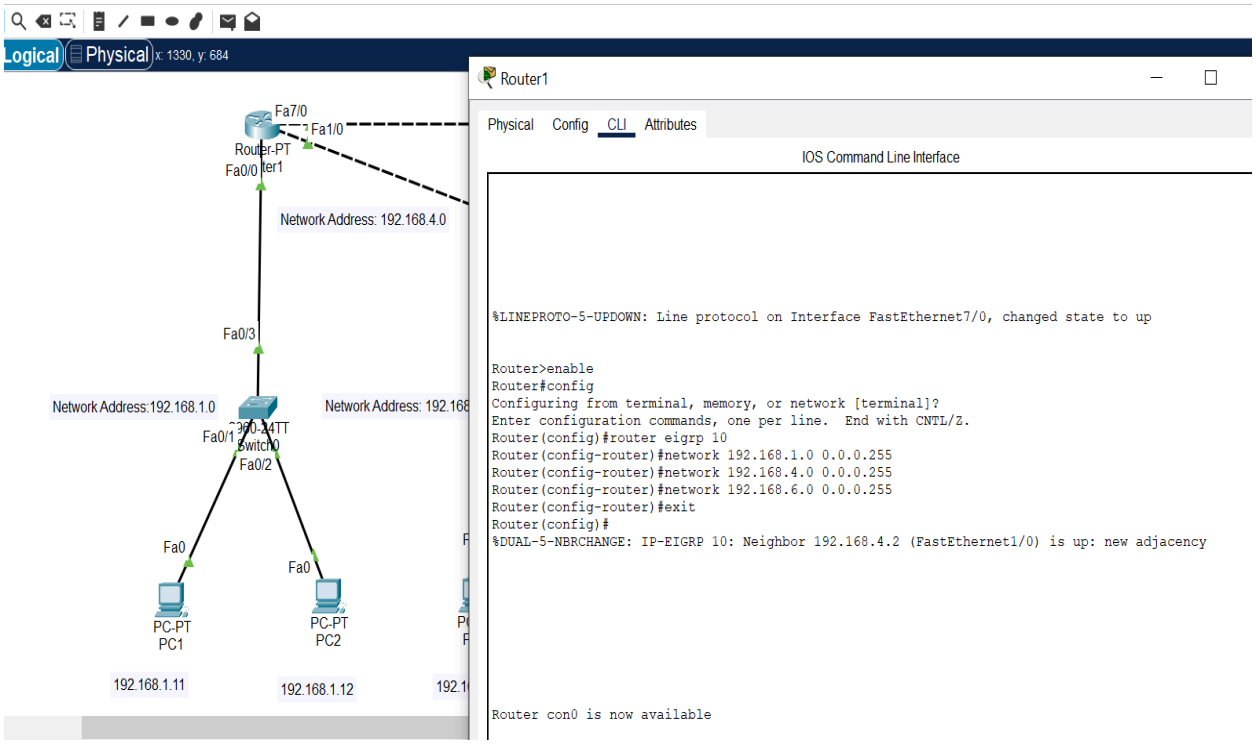
router eigrp 10

network 192.168.1.0 0.0.0.255

network 192.168.4.0 0.0.0.255

network 192.168.6.0 0.0.0.255

exit



Router 2:

Router>enable

Router#config

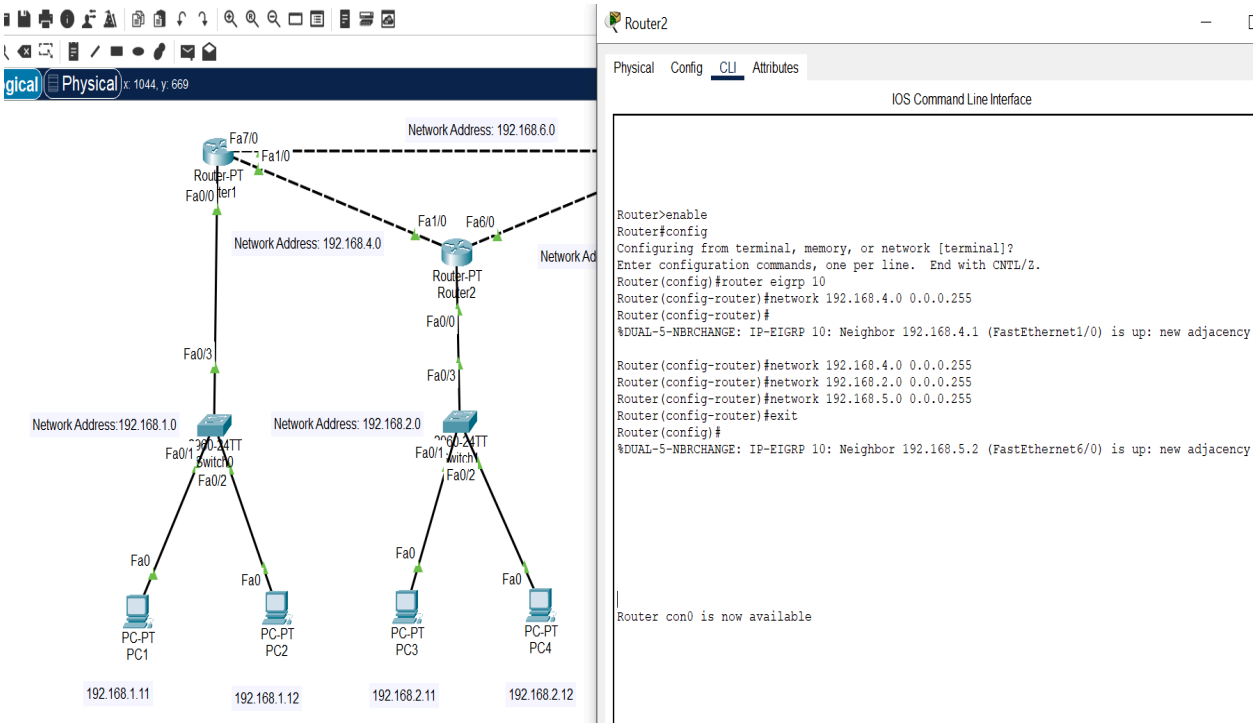
router eigrp 10

network 192.168.4.0 0.0.0.255

network 192.168.2.0 0.0.0.255

network 192.168.5.0 0.0.0.255

exit



Router 3:

Router>enable

Router#config

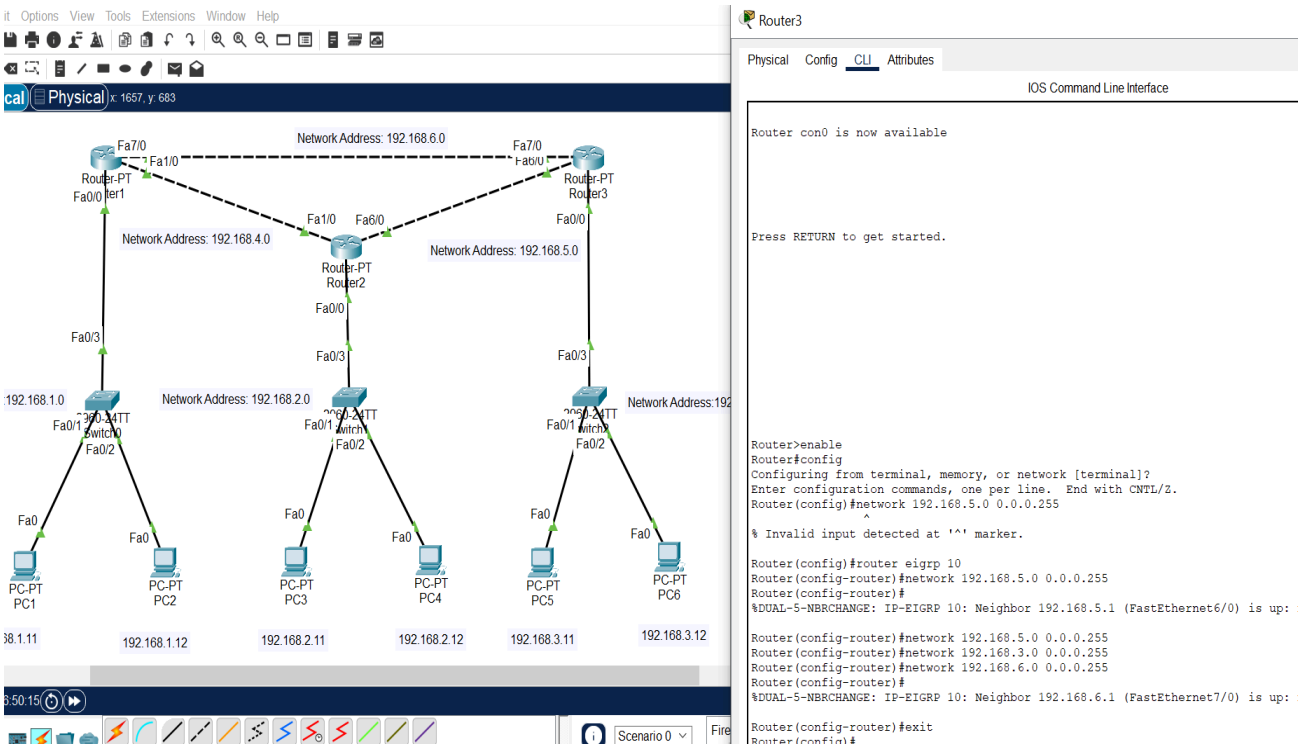
router eigrp 10

network 192.168.5.0 0.0.0.255

network 192.168.3.0 0.0.0.255

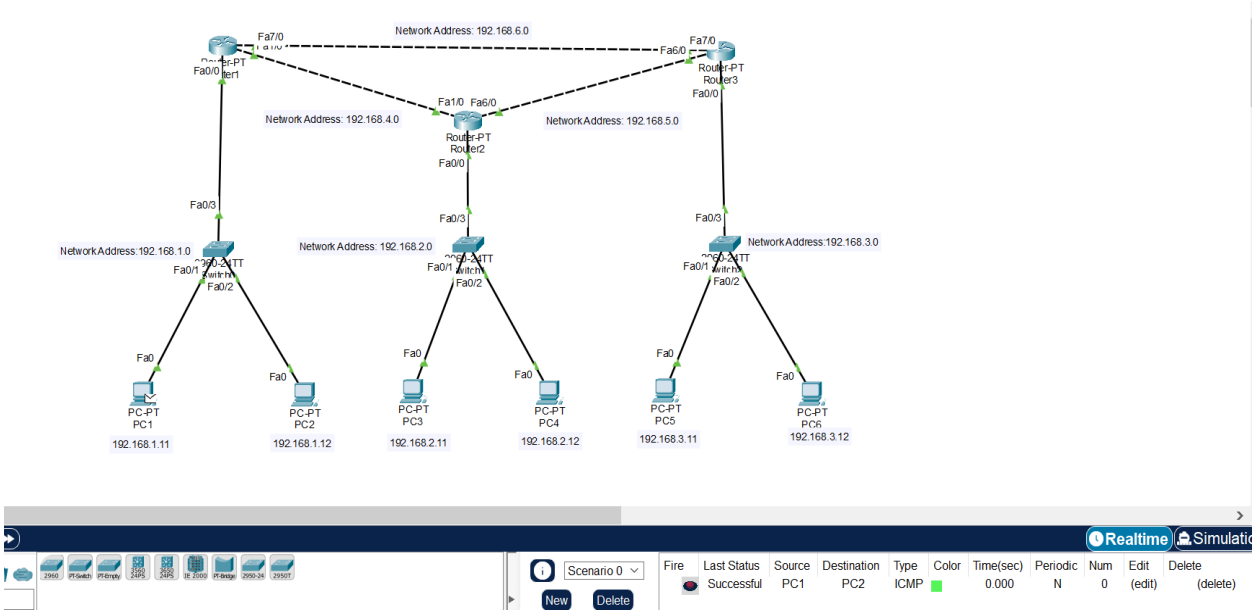
network 192.168.6.0 0.0.0.255

exit

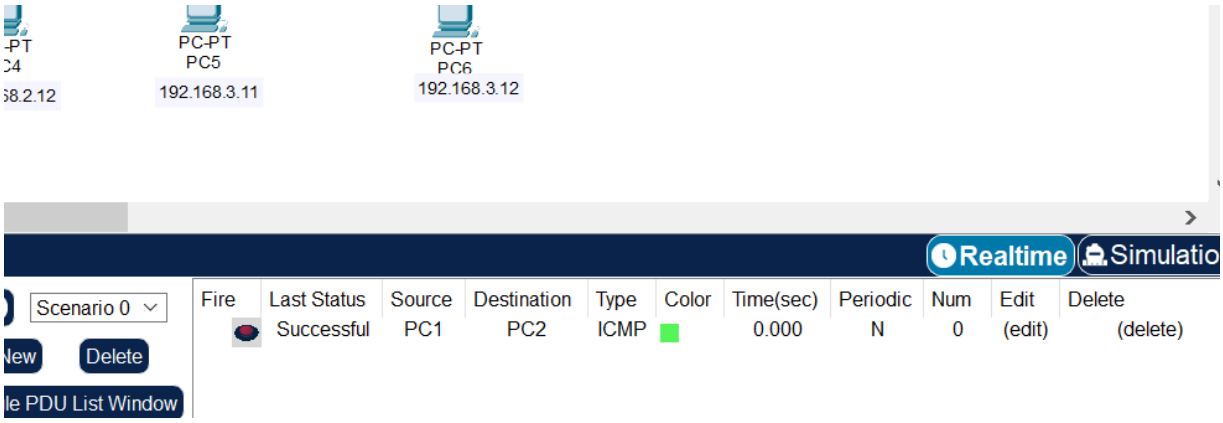


3.Experimental Result:

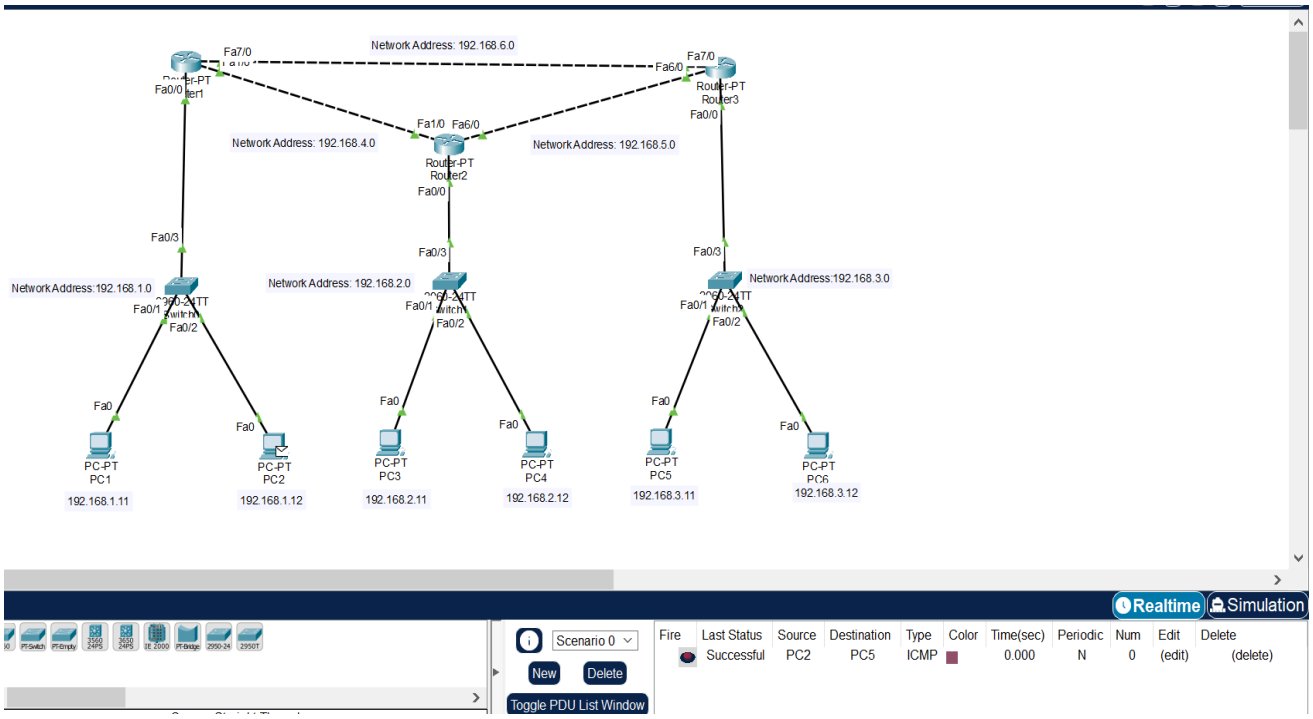
➔ **Sending data packets from PC1 to PC2 within the same router network:**



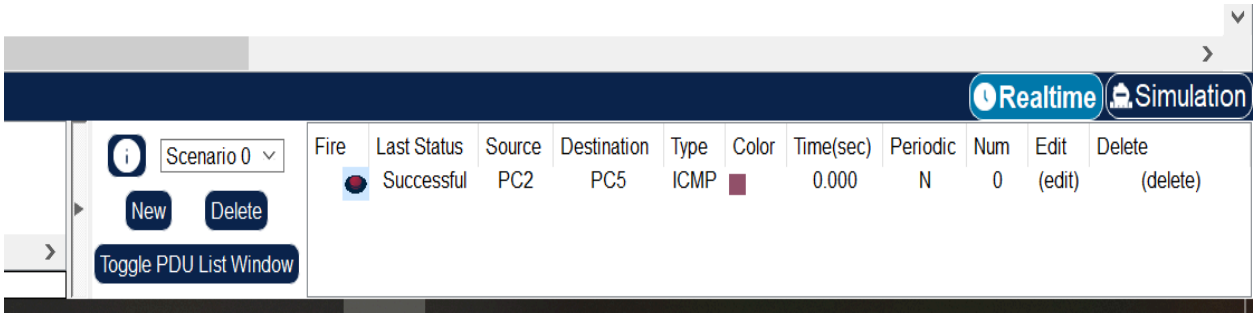
➔ **Packet successfully sent from PC1 to PC2.**



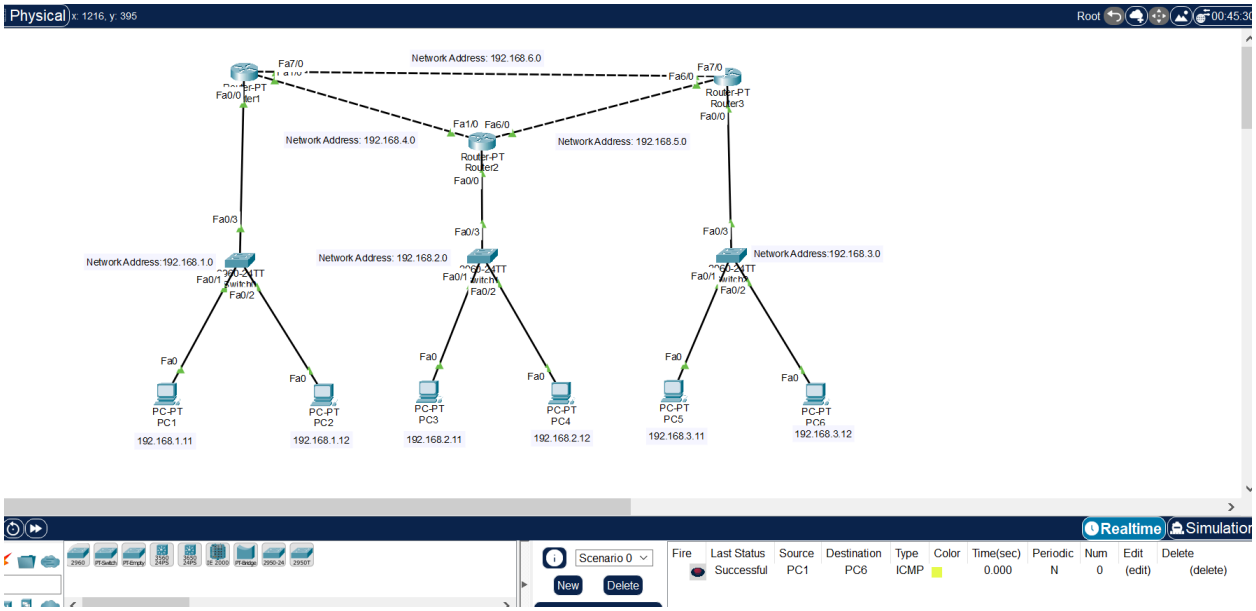
➔ Sending data packets from PC2 to PC5 on different routers:



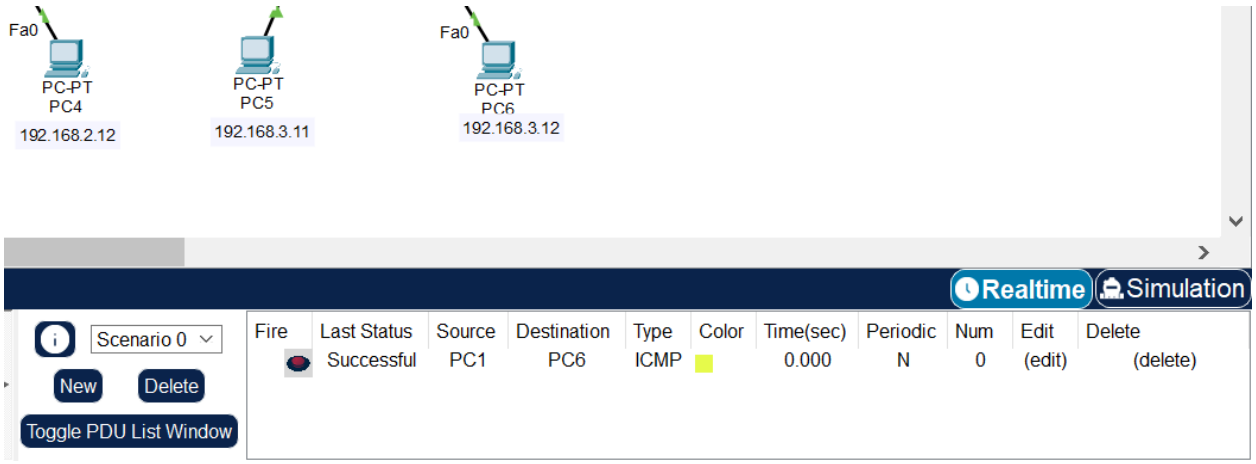
➔ Packet successfully sent from PC2 to PC5.



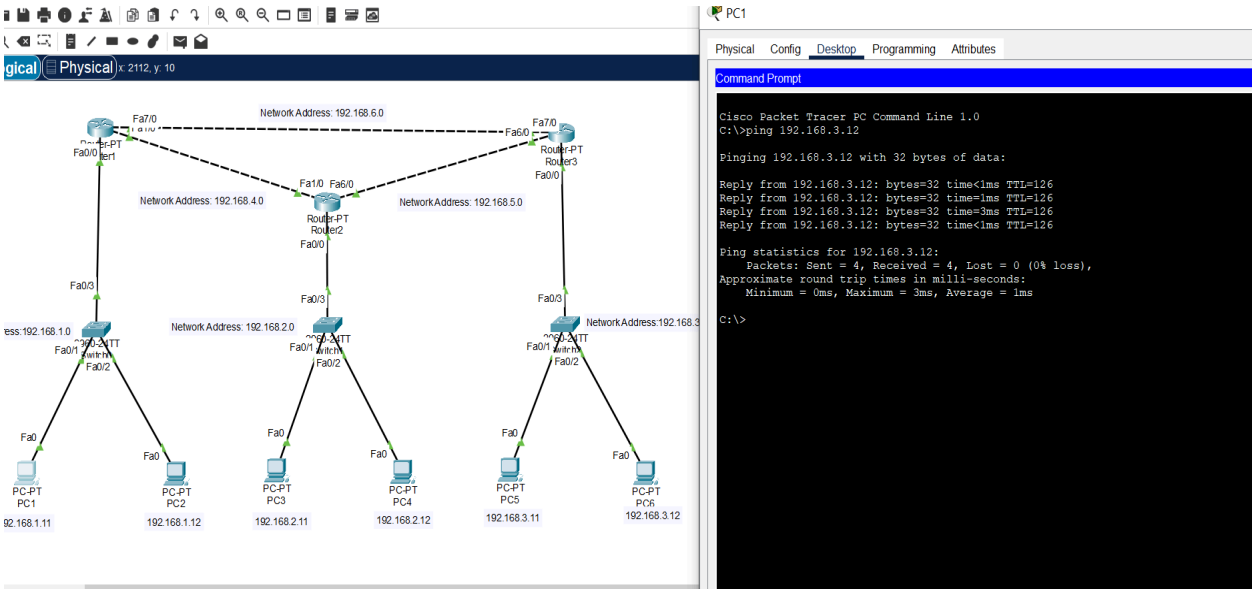
➔ Sending data packets from PC1 to PC6 on different routers:



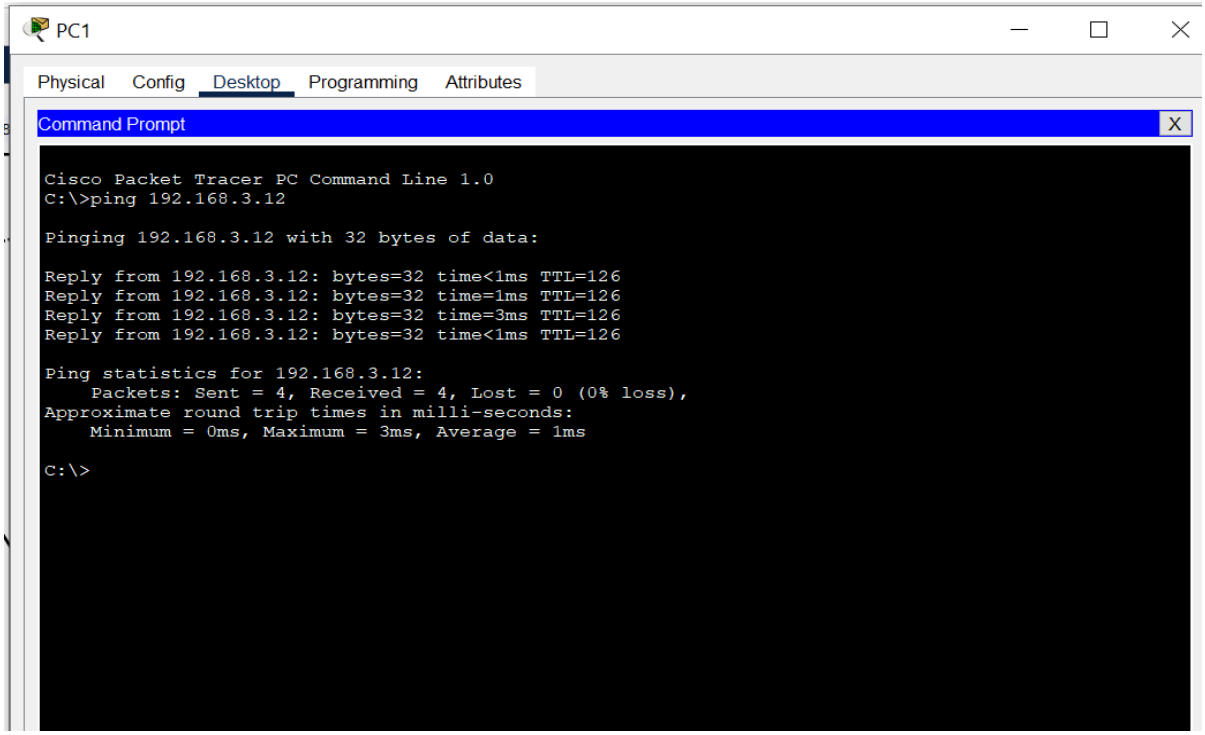
➔ Packet successfully sent from PC1 to PC6.



- ☐ **PING operation between PC1 on the first router and PC6 on the third router:**



➔ **Ping successfully sent from PC1 to PC6.**



4.Conclusion

We have successfully established a network consisting of six PCs, three switches, and three routers. Each device in the network has been assigned a Class C IP address, and EIGRP has been implemented as the networking protocol to facilitate efficient routing and data communication.After configuring the network, we tested the connectivity between PC1 on the first router and PC6 on the third router using the PING command. The successful response, with an average round-trip time of 1ms, confirms that the network is functioning correctly and supports seamless communication across multiple routers.This project demonstrates that the network design is both practical and effective, making it suitable for a small to medium-sized company or organization with various departments or locations. The hierarchical network structure, combined with the use of EIGRP, ensures robust and reliable connectivity, meeting the data communication needs of the organization.